1. Proof and Mathematical Logic

Logic and Notation

1. Write the following statements as a conditional statement in the form $p \implies q$.
   (a) "A monkey is happy only if he is eating a banana."
   (b) "A snake will not bite you provided you don't step on its tail."
   (c) "A donkey laughs whenever he sees a mule."
   (d) "Happiness is a necessary condition for Wealth."
   (e) "Happiness is a sufficient condition for Wealth."

2. Indicate whether each statement is True or False.
   (a) Jupiter is a planet and Neptune is a moon.
   (b) Jupiter is a planet or Neptune is a moon.
   (c) Elvis was a woman or Cleopatra was a man.
   (d) *TikTok* is a social media app, or *Lord of the Rings* was written by J.R.R. Tolkien.
   (e) If the capital of Egypt is Cairo, then apples can be used to make cider.
   (f) If Napolean was born in Zimbabwe, then the eigenstates of the quantum harmonic oscillator are proportional to Hermite polynomials.
   (g) If the dodo is extinct, then pigs can fly!
   (h) It is not the case that if Luke Skywalker was a Jedi, then his father was not Darth Vader.

3. Translate the following into mathematical notation.
   (a) Six is not prime or eleven is not prime.
   (b) The square of 10 is 50 and the cube of 5 is 12.
   (c) If 7 is an integer then 6 is not an integer.
   (d) If both 2 and 5 are prime then 2×5 is not prime.

   Which of these are true, which are false, and which are neither?

4. Construct truth tables for the following statements.
   (a) $(p \wedge q) \vee (\sim p \wedge \sim q)$
   (b) $[\sim q \wedge (p \implies q)] \implies \sim p$
   (c) $[(p \vee q) \wedge r] \implies (p \wedge r)$

   Which of the statements above is logically equivalent to a connective we have studied? Which of the statements above is a tautology?

5. Translate the following into mathematical notation
   (a) All rational numbers are larger than 6.
   (b) There is a real number solution to $x^2 + 3x - 7 = 0$.
   (c) There is a natural number whose cube is 8.

   Which of these are true, which are false, and which are neither?

6. Translate the following mathematical statements into English.
   (a) $(\forall a \in \mathbb{Q})\ a + 0 = a$
   (b) $(\forall x \in \mathbb{R})\ x^2 > 1$

7. The following two statements look similar, but say very different things. Which is true, and which is false?

(a) $(\exists b \in \mathbb{Z})\ [(\forall a \in \mathbb{Z})\ \ a + b = 0]$

(b) $(\forall a \in \mathbb{Z})\ [(\exists b \in \mathbb{Z})\ \ a + b = 0]$

8. Find the negation of

(a) $(\forall x \in \mathbb{R})\ \ x^2 = 10$

(b) $(\exists y \in \mathbb{N})\ \ y < 0$

(c) $(\exists a \in \mathbb{N})\ [(\forall x \in \mathbb{R})\ \ ax = 4]$

(d) $(\forall y \in \mathbb{Q})\ [(\exists x \in \mathbb{R})\ \ x/y = 30]$

9. Verify the following:

(a) $p \wedge q \equiv q \wedge p$

(b) $\sim (p \implies q) \equiv (p \wedge \sim q)$

What can you conclude about (i) $p \wedge q \iff q \wedge p$ and (ii) $\sim (p \implies q) \iff (p \wedge \sim q)$?

Counter Examples

10. Use a counterexample to show that the following statements are false.

(a) If the product of two integers is even then both of those integers are even.

(b) For all real numbers, if $x^2 = y^2$ then $x = y$.

(c) Let $a \in \mathbb{Z}$. If a divided by 7 gives remainder of 4, then $5a$ divided by 7 gives remainder of 4.

Direct Proofs

11. Prove the following results for the integers.

(a) The product of an even integer with an odd integer is even.

(b) The sum of an even integer and an odd integer is odd.

(c) The cube of an odd integer is odd.

(d) Let $a \in \mathbb{Z}$. If a divided by 7 gives remainder of 4, then $15a$ divided by 7 gives remainder of 4.

(e) If k is odd, then $k^2 - 1$ is divisible by 4.

Domains

12. For each equation below, list the sets of numbers $(\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R})$ in which there exists a solution.

(a) $x^2 - 8 = 0$

(b) $x + 8 = 0$

(c) $4x = 8$

(d) $x^2 + 8 = 0$

(e) $3x + 8 = 0$

(f) $0x = 8$

Contrapositive

13. Prove the following theorems using the contrapositive.

(a) Let $n \in \mathbb{Z}$. Prove that if n^4 is even, then n is even.

(b) Let $n \in \mathbb{Z}$. Prove that if n^3 is odd, then n is odd.

(c) For all $m, n \in \mathbb{Z}$, if mn is odd then m and n are odd.

Proof by Contradiction

14. Consider the following theorem:

Let $p, q \in \mathbb{Z}$ with $p, q > 0$. If $pq = 1$ then $p = q = 1$.

(a) Begin a proof by contradiction by assuming that at least one of p and q is not equal to 1.

(b) Starting with $pq = 1$, arrive at a contradiction.

15. Consider the following equation:
$$x^2 - n^2y^2 = 1$$
where $n \in \mathbb{N}$ is fixed.

Show that it has no positive integer solutions for x, y using a proof by contradiction in the following steps.

(a) Begin by assuming that there exists a positive integer solution.

(b) Factorized the left-hand side, and conclude that both factors are integers.

(c) Using the result of Question 14, conclude both factors equal 1.

(d) Attempt to solve the two equations you have just developed and arrive at a contradiction.

16. *The Adventures of π-casso, the Mathematical Artist*
Your friend π-casso has invited you over to see their new artwork. Before they unveil it, they tell you that it is a 3×3 grid of squares, each painted either red or blue. They go on to say that they used a special rule to paint it:

Every square has either 2 or 4 blue neighbours.

(a) Suppose that they tell you the centre square is red. Prove that the remaining squares are all blue using a contradiction argument. Hint: You will need to consider 2 cases.

(b) Suppose that they tell you the centre square is blue. Prove that at least one of the remaining squares is red using a contradiction argument.

(c) How many possible paintings are there that satisfy the special rule?

(d) *Extension.* Consider the paintings of 4×4 grids that use the same special rule. What are the possibilities?

More Proofs

17. Prove the following theorems by dividing into two or more cases.

(a) Let $n \in \mathbb{Z}$. Prove that if n is not divisible by 3, then n^2 is not divisible by 3.

(b) For all $m, n \in \mathbb{Z}$, if m and n are either both odd or both even then $m + n$ is even.

Irrational numbers

18. Prove that the following are irrational:

(a) $\sqrt{3}$ (b) $\sqrt{15} + \sqrt{5}$ (c) $\log_2 7$

Mathematical Induction

19. Prove by induction that each formula is true for every integer $n \geq 1$.

(a) $2 + 7 + 12 + \cdots + (5n - 3) = \frac{1}{2}n(5n - 1)$

(b) $1 + 2 \cdot 2 + 3 \cdot 2^2 + 4 \cdot 2^3 + \cdots + n \cdot 2^{n-1} = 1 + (n - 1)\, 2^n$

(c) $\dfrac{1}{1 \cdot 2} + \dfrac{1}{2 \cdot 3} + \dfrac{1}{3 \cdot 4} + \cdots + \dfrac{1}{n(n+1)} = \dfrac{n}{n+1}$

(d) $\dfrac{a^{n+1} - b^{n+1}}{a - b} = a^n + a^{n-1}b + a^{n-2}b^2 + \cdots + ab^{n-1} + b^n \quad (a \neq b)$

Begin by rewriting these equations using summation notation.

20. Prove by induction that the following statements are true for every natural number n:

(a) 3 is a factor of $n^3 - n + 3$;

(b) 9 is a factor of $10^{n+1} + 3 \cdot 10^n + 5$;

(c) 4 is a factor of $5^n - 1$;

(d) $x - y$ is a factor of $x^n - y^n$;

(e) $7^{2n} - 48n - 1$ is divisible by 2304 (for all $n \geq 1$).

21. Write the following inequalities in summation notation and then prove them for all $n \in \mathbb{N}$, using summation notation throughout your proof. If you find this difficult, first try the proofs using more informal notation.

    (a) $1^3 + 2^3 + \cdots + (n-1)^3 < \frac{1}{4}n^4 < 1^3 + 2^3 + \cdots + n^3$ for $n \geq 2$;

    (b) $1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{n}} \geq \sqrt{n}$.

22. In each case try to find $n_0 \in \mathbb{N}$ such that the inequality appears to hold true for $n \geq n_0$. If you think you have found such an n_0, give a proof by induction that the inequality holds true for $n \geq n_0$. If you think that no such n_0 exists, try to prove that it cannot exist.

    (a) $1 + 2n \leq 3^n$

    (b) $n! > 2^n$

    (c) $n(n+1) \geq (2n-1)^2$

    (d) $n! > 2n^3$

23. If $\mathbf{A}$ is a square matrix and $\mathbf{1}$ is the identity matrix (of the same size), we define $\mathbf{A}^n$ by

$$\mathbf{A}^0 = \mathbf{1} \quad \text{and} \quad \mathbf{A}^{n+1} = \mathbf{A}^n\mathbf{A} \text{ for } n = 0, 1, 2, \ldots,$$

which is a more precise and careful way of saying that

$$\mathbf{A}^n = \underbrace{\mathbf{A}\,\mathbf{A}\cdots\mathbf{A}}_{n \text{ factors}}.$$

Prove by mathematical induction that for all natural numbers n

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & 0 \\ n & 1 \end{pmatrix}.$$

2. Set Theory

Sets

For questions 1 to 3, let $A = \{1, 2, 3, 4\}$, $B = \{1, 3, 5, 7\}$ and $C = \{2, 3\}$.

1. Find the following sets:

   (a) $A \cup B$
   (b) $A \cap B$
   (c) $A \cup C$
   (d) $A \cap C$
   (e) $A \cup B \cup C$
   (f) $A \cap B \cap C$

2. Find the following sets:

   (a) $B \times C$
   (b) $A \times \{\}$

3. Which of the following statements are true?

   (a) $3 \subseteq B$
   (b) $\{\} \subseteq A$
   (c) $7 \in B \cup C$
   (d) $C \subseteq B$
   (e) $C = B \cup C$
   (f) $C \in A$
   (g) $(\forall a \in A)\ a \leq 4$
   (h) $(\forall b \in B)[(\exists c \in C)\ b - c \geq 0]$

4. Let A, B and C be sets. Let $\mathcal{U} = A \cup B \cup C$. Prove the following statements.

   (Please write your proof using the definitions of $=$, $\subseteq$, $\cup$, $\cap$, etc. from the lectures on set theory. Do not use set algebra.)

   (a) $A \subseteq [(A \cap B) \cup (A \cap (\mathcal{U} \backslash B))]$
   (b) $A \subseteq B \implies (C \cap A) \subseteq (C \cap B)$
   (c) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
   (d) $A \subseteq B$ if and only if $A \cup B \subseteq B$

5. *The Cantor Set*

   We are going to construct an unusual set made famous by Georg Cantor.

   We begin with the set $C_0 = [0, 1]$

   (a) Draw this set on the real number line. What is its length?
   (b) Construct C_1 by removing the middle third of C_0 as an open interval. In other words, remove $(\frac{1}{3}, \frac{2}{3})$. Write C_1 as the union of two intervals. Draw C_1 on the real line, and determine its length.
   (c) Construct C_2 by removing the middle third of each segment in C_1. C_2 should have 4 segments. Write C_2 as a union of 4 intervals. Draw it on the real line, and determine its length.
   (d) By now you probably understand the pattern. Draw a few more: C_3, C_4, C_5 etc.
   (e) The Cantor Set C is what remains after you repeat this process *ad infinitum*. Find several points that are in the set. Find several points that aren't in the set.
   (f) The Cantor Set is a simple example of a *fractal*. It exhibits self-similairty: meaning if you zoom in on sections, you see the overal shape repeating itself. Type "fractal cauliflower" into Google to see another example of a fractal.

Supremum and Infimum

6. Find the supremum and infinum of the set S (where $S \subseteq \mathbb{R}$), if they exist, and if they do, state whether the supremum or infimum is an element of S. You do not need to prove your answer is correct.

   (a) $S = \{x \mid x^2 \leq 9\}$
   (b) $S = \{x \mid |x-2| < 3\}$
   (c) $S = \{x \mid |2x+1| < 5\}$
   (d) $S = \{x \mid |x-2| < 3 \wedge |x+1| < 1\}$
   (e) $S = \{x \mid |x+2| \leq 2 \vee |x| > 1\}$
   (f) $S = \{x \in \mathbb{Q} \mid x^2 \leq 7\}$

7. Let $S \subseteq \mathbb{R}$ and $c \in \mathbb{R}$. Define $c + S = \{c + x \mid x \in S\}$ and $cS = \{cx \mid x \in S\}$. If S is bounded above and below, prove the following:

   (a) $c + S$ and cS are both bounded above and below
   (b) $\sup(c + S) = c + \sup(S)$
   (c) $\sup(cS) = c\sup(S)$ for $c \geq 0$
   (d) $\sup(cS) = c\inf(S)$ for $c \leq 0$

8. Let $A \subseteq \mathbb{R}$. Prove that the supremum of A is unique if it exists.

   Hint: A useful method for showing some quantity is unique is to assume there are two, and prove they must be equal. In this case, assume that A has two suprema, s_1 and s_2, and show that $s_1 = s_2$. If $s_1 \leq s_2$ and $s_2 \leq s_1$, then $s_1 = s_2$

Inequalities

This subject will involve heavy use of inequalities. The following exercises will help you practice and become comfortable manipulating them.

Recall the following:

The Triangle Inequality: For all $a, b \in \mathbb{R}$ we have $|a + b| \leq |a| + |b|$.

9. Let $a, b, c \in \mathbb{R}$ and let $a < b$. Which of the following statements are always true? Which are sometimes true? Which are never true?

   (a) $a + 1 < b + 1$
   (b) $a + c < b + c$
   (c) $5a < 5b$
   (d) $ac < bc$
   (e) $\dfrac{1}{a} < \dfrac{1}{b}$
   (f) $\dfrac{1}{c+a} < \dfrac{1}{c}$
   (g) $c < c + a$
   (h) $-a < -b$

10. Express the solutions to the following inequalities as intervals of $\mathbb{R}$.

    (a) $|1 + 2x| \leq 4$
    (b) $|x + 2| \geq 5$
    (c) $|x - 5| < |x + 1|$
    (d) $|x - 2| < 3 \vee |x + 1| < 1$
    (e) $|x - 2| < 3 \wedge |x + 1| < 1$

11. Use the triangle inequality or other properties of inequalities to find a bound for $|f|$ on the stated interval.

    (a) $f(x) = \dfrac{2x^2 + 1}{x + 3}, \quad |x| < 1$
    (b) $f(x) = \dfrac{x^3 + 3x + 1}{10 - x^3}, \quad |x + 1| < 2$

12. Prove that $\forall a, b \in \mathbb{R}$ that $|a - b| \geq ||a| - |b||$.

13. Let $a, b \in \mathbb{R}$. If $0 < \epsilon < \min\{|a|, |b|\}$ show that

$$\left|\frac{a+\varepsilon}{b+\varepsilon}\right| \leq \frac{|a|+\varepsilon}{|b|-\varepsilon}$$

14. Let $a, b \geq 0$, $p > 1$ and $q = p/(p-1)$. We will prove that

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

(a) Consider the cases where either $a = 0$ or $b = 0$: the inequality is trivially true for these.

(b) Treat a as a real variable and define

$$f(x) = \frac{x^p}{p} + \frac{b^q}{q} - bx, \quad x > 0$$

Show that f has a minimum at $x = b^{\frac{1}{p-1}}$ using calculus techniques.

(c) Find the value of the function at this point, and conclude $f(x) \geq 0$.

(d) Finish off the proof.

3. The Real Numbers

Proofs with the Real Number Axioms

1. Let x, y, $z \in \mathbb{R}$. Using only the axioms of the real numbers, prove the following obvious results.

   (a) If $x + z = y + z$, then $x = y$.

   (b) $0x = 0$

   (c) $-(-x) = x$

   (d) If $xy = 0$, then $x = 0$ or $y = 0$.

   (e) If $x < y$ and $z < 0$ then $xz > yz$.

   (f) If $x > 0$, then $1/x > 0$.

Properties of Number Sets

2. What property separates the following sets? In other words, what vital property does one set have that the other doesn't?

   (a) $\mathbb{N}$ and $\mathbb{Z}$

   (b) $\mathbb{Z}$ and $\mathbb{Q}$

   (c) $\mathbb{Q}$ and $\mathbb{R}$

The Archimedean Principle

3. Give a proof of the following statement

$$(\forall \epsilon > 0) \ \ [(\exists n \in \mathbb{N})\ 1/n < \epsilon]$$

Existence of the square root of 2

4. Throughout the course we have talked about the number $\sqrt{2}$. However, we have never actually shown that it follows from the Axioms of the Real Numbers that square roots exist. In this question, we will prove that there exists a positive real number α such that $\alpha^2 = 2$.

   Throughout this exercise, only use algebraic facts which readily follow from the Axioms of the Real Numbers. Be careful to *not* assume the existence of square roots.

   (a) Let $A = \{x \in \mathbb{R} : x^2 \leq 2\}$. Show A is bounded above in $\mathbb{R}$. That is, select a specific real number β which is an upper bound for A. Then prove β is indeed an upper bound for A in $\mathbb{R}$.

   (b) Explain why the supremum α of the set A exists in $\mathbb{R}$.

   (c) Suppose that $\alpha^2 < 2$.

      (i) Use the Archimedean Principle to show that there exists $n \in \mathbb{N}^+$ such that $(\alpha + 1/n)^2 < 2$.

      (ii) Using Part (c)(i), show that α is not an upper bound of A. (This of course contradicts Part (b)!)

   (d) Suppose that $\alpha^2 > 2$.

      (i) Use the Archimedean Principle to show that there exists $n \in \mathbb{N}^+$ such that $(\alpha - 1/n)^2 > 2$.

      (ii) Using Part (d)(i), show that α is not the least upper bound of A. (This again contradicts Part (b)!)

   (e) Explain why $\alpha > 0$ and $\alpha^2 = 2$.

4. Functions and Sequences

ϵ–M Definition of Convergence

1. Use ϵ-M definition of convergence to prove the following.

   (a) $\dfrac{1}{n+2} \to 0$ (b) $\dfrac{1}{(n+4)^2} \to 0$ (c) $\dfrac{n^2}{2n^2+4} \to \dfrac{1}{2}$.

2. For the following sequences, guess the limit, then verify the result using an ϵ–M argument.

   (a) $\dfrac{1}{n^2+1}$ (b) $\dfrac{2n}{n+1}$ (c) $\dfrac{3n}{2n+5}$ (d) $\dfrac{n^2}{2n^2+3}$

3. Without appealing to Algebra of Limits Theorem prove the following:

   If $u_n \to L$, then $2u_n \to 2L$.

4. Using the Algebra of Limits Theorem, prove the following statement

   If $\lim_{n\to\infty} a_n = L$, $L \neq 0$ and a_n is never equal to 0, then $\lim_{n\to\infty} 1/a_n = 1/L$.

5. Using the $\epsilon - M$ definition of convergence, prove the Sandwich Theorem:

   Let a_n, b_n and c_n be sequences. If $a_n \leq b_n \leq c_n$, $a_n \to L, c_n \to L$, then $b_n \to L$.

Subsequences

6. Prove that the following sequences diverge.

   (a) $e_n = \dfrac{n+1}{\sqrt{n}}$

   (b) $g_n = 1 + (-1)^n$

7. For each of the following sequences, find all r such that a subsequence converges to r. Make a conclusion about the convergence or otherwise of the sequence.

   (a) $a_n = (-1)^n \dfrac{n}{n+1}$ (b) $b_n = \sin\left(\dfrac{n\pi}{2}\right) \cdot \dfrac{n}{n+1}$ (c) $c_n = (-1)^{n+1} \dfrac{n^2}{n^3+5}$

8. Let $k \in \mathbb{R}$. Using the $\epsilon - M$ definition of convergence, show that if $b_n \to \beta$, then $kb_n \to k\beta$.

9. Let (a_n) be a sequence such that the subsequences (a_{2n}) of even terms and (a_{2n-1}) of odd terms converge to the same limit L. Show that (a_n) converges to L.

Monotone Sequences

10. For each sequence, classify it as monotone increasing, monotone decreasing, neither, or both.

    (a) $(\sqrt{n})$

    (b) $\left(\frac{(-1)^n}{n}\right)$

    (c) $\left(n + \frac{1}{n}\right)$

    (d) $(3, 3, 3, 3 \ldots)$

    (e) $\left(\frac{1}{n}\right)$

11. For the sequences in the previous part that are monotone, determine if they are convergent.

12. Let (a_n) be a sequence that is both monotone increasing and decreasing. Show $a_n \to a_1$.

Cauchy Sequences

13. Use the definition of Cauchy, show that the sequence given by $f_n = 4 + \frac{(-1)^n}{3n}$ is Cauchy

14. Use the definition of a Cauchy to show that the sequence given by $f_n = \dfrac{n+3}{2n-1}$ is Cauchy.

Contractive Sequences

15. Consider the sequence (f_n) defined inductively by $f_1 = 1$ and, for all $n \geq 1$, $f_{n+1} = 1/(1 + f_n)$.
    (a) Using induction, prove that $1/2 \leq f_n \leq 1$ for all $n \geq 1$.
    (b) Show that $|f_{n+2} - f_{n+1}| \leq \frac{4}{9} \cdot |f_{n+1} - f_n|$ for all $n \geq 1$.
    (c) Using the previous part and results from the lectures, show that (f_n) converges. Compute the limit.
16. Consider the cubic equation $x^3 - 6x + 3 = 0$. Find a contractive sequence that converges to a solution $L \in (0, 1)$ of this equation.

Existence

17. For each item below, give an example or show it doesn't exist:
    (a) A bounded sequence with an unbounded subsequence.
    (b) An unbounded sequence with a bounded subsequence.
    (c) A monotone sequence that is not Cauchy.
    (d) A bounded sequence that not Cauchy.
    (e) A bounded monotone sequence that is not Cauchy.
    (f) A Cauchy sequence that is not bounded.
    (g) A Cauchy sequence that is not monotone.
    (h) A Cauchy sequence that has an unbounded subsequence.
    (i) An unbounded sequence with a Cauchy subsequence.
    (j) A bounded sequence with a divergent monotone subsequence.
    (k) A divergent sequence with both an monotone increasing subsequence and a monotone decreasing subsequence.
    (l) A convergent sequence (f_n) for which $f_n \not\to \sup\{f_n \mid n \in \mathbb{N}^+\}$ and $f_n \not\to \inf\{f_n \mid n \in \mathbb{N}^+\}$.
    (m) A convergent monotone sequence (f_n) for which $f_n \not\to \sup\{f_n \mid n \in \mathbb{N}^+\}$ and $f_n \not\to \inf\{f_n \mid n \in \mathbb{N}^+\}$.

5. Limits and Continuity

Limit Points

1. Find all the limit points of the following sets:

   (a) $E = \left\{ \frac{1}{n} \,\middle|\, n \in \mathbb{N} \right\}$

   (b) $E = \left\{ \frac{1}{m} + \frac{1}{n} \,\middle|\, m, n \in \mathbb{N} \right\}$

   (c) $E = \left\{ (-1)^n \left(1 + \frac{1}{n}\right) \,\middle|\, n \in \mathbb{N} \right\}$

   (d) $E = \mathbb{Z}$

   (e) $E = \mathbb{Q}$

2. Let's prove the following theorem:

   THEOREM:
   Let $E \subseteq \mathbb{R}$ and let $a \in \mathbb{R}$. a is a limit point of E if and only if there exists a sequence of points (x_n) such that $x_n \in E$ and $x_n \neq a$ for all $n \in \mathbb{N}^+$ and $x_n \to a$ as $n \to \infty$.

   (a) First let's show the "if" part. Suppose that there exists sequence of points (x_n) such that $x_n \in E$ and $x_n \neq a$ for all $n \in \mathbb{N}^+$ and $x_n \to a$ as $n \to \infty$. Using the definition of limit point and $\epsilon - M$ definition of convergence, show that a is a limit point of E.

   (b) Next let's show the "only if" part. Suppose that a is a limit point of E.

      (i) Use the definition of limit point to explain why for each $n \in \mathbb{N}^+$ there exists $x_n \in E$ such that $x_n \neq a$ and $|x_n - a| < 1/n$.

      (ii) Using Part (i) and the $\epsilon - M$ definition of convergence, show that $x_n \to a$.

ϵ-δ Limits

3. Using the ϵ-δ definition of the limit, prove the following limits:

   (a) $\lim_{x \to 3} 5x = 15$

   (b) $\lim_{x \to 3} (5x + 2) = 17$

   (c) $\lim_{x \to 2} \frac{3x^2 - 12}{x - 2} = 12$

   (d) $\lim_{x \to 3} \frac{3}{x} = 1$. (This one is tricky! Notice that $\left| \frac{3}{x} - 1 \right| < \frac{|x-3|}{|x|}$. What can you say about $\frac{1}{|x|}$ when $|x-3| < 1$?)

Limits to Infinity

4. Prove the following limits using the definition of limits to ∞:

   (a) $\lim_{x \to \infty} c = c$ for each $c \in \mathbb{R}$.

   (b) $\lim_{x \to \infty} \frac{1}{x^p} = 0$ for each $p > 0$.

5. Prove $\lim_{x \to \infty} \left(4 + \frac{1}{\sqrt{x}}\right) = 4$.

6. Prove $\lim_{x \to \infty} \sqrt{x} = \infty$.

7. Let $f(x)$ be a function such that $\lim_{x \to \infty} f(x) = \infty$. Define the sequence (f_n) by $f_n = f(n)$ for all $n \in \mathbb{N}^+$. Prove

$$\lim_{x \to \infty} f_n = \infty.$$

Weird Functions

8. Consider the following function:

$$f(x) = \begin{cases} 0 & x \in \mathbb{Q} \\ 1 & x \notin \mathbb{Q} \end{cases}$$

Using an ϵ-δ argument, prove that $f(x)$ diverges as x approaches $\sqrt{2}$.

9. Consider the following function:

$$f(x) = \begin{cases} 0 & x \in \mathbb{Q} \\ x & x \notin \mathbb{Q} \end{cases}$$

Using an ϵ-δ argument, prove that $f(x)$ converges as x approaches 0.

10. (Challenge) Recall that each non-zero rational number $x \in \mathbb{Q}$ can be expressed as $x = p/q$ where p, q are relatively prime integers and $q > 0$. Consider Thomae's function:

$$f(x) = \begin{cases} 0 & x \notin \mathbb{Q} \\ 1 & x = 0 \\ 1/q & x \neq 0 \text{ and } x = p/q \in \mathbb{Q} \text{ (as above)} \end{cases}$$

   (i) Show that for each $a \in \mathbb{R}$, $f(x)$ converges as x approaches a.

   (ii) Let $a \in \mathbb{R}$. Show that $f(x)$ is continuous at a if and only if $a \notin \mathbb{Q}$.

Continuity

11. Using ϵ-δ arguments prove that if two functions f, g are continuous at a, then so are

   (a) $f - g$

   (b) $f + 2g$

12. Determine if the function f is continuous on the interval $[-1, 1]$.

   (a) $f(x) = |x|$

   (b) $f(x) = \begin{cases} (1-x)^{-1}, & x \neq 1; \\ 1, & x = 1. \end{cases}$

   (c) $f(x) = \begin{cases} \dfrac{|x|}{x}, & x \neq 0; \\ 1, & x = 0. \end{cases}$

13. Let $f : I \to \mathbb{R}$ be continuous on the interval I. Prove the function g defined by $g(x) = |f(x)|$ is continuous on I. Give a counterexample that proves that the converse is false?

The Intermediate Value Theorem

14. (a) Let $f(x) = x^3 - 5x^2 + 7x - 9$. Prove that there is a real number c such that $f(c) = 100$.

   (b) Let $f(x) = x^5 - 3x^4 - 2x^3 - x + 1$. Proof that f has at least one root between 0 and 1.

   (c) Show that equation $x + \sin x = 1$ has at least one solution in the interval $[0, \pi/6]$.

   (d) Let $f : [0, 1] \to [0, 1]$ be a continuous function. Prove that $\exists c \in [0, 1]$ such that $f(c) = c$.

   Hint: Define a function g so that $g(x) = 0$ exactly when $f(c) = c$.

   (e) Let $f : [0, 1] \to \mathbb{R}$ be a continuous function satisfying $f(0) = f(1) = 0$. Prove that $\exists c \in [0, 1/2)$ such that $f(c + 1/2) = f(c)$.

6. Differentiation and Integration

The Definition of the Derivative

1. In this exercise, we will prove the *power rule* for computing derivatives. We will proceed by induction.
   (a) Using the definition of differentiable at c, prove that if $f(x) = x$ then $f'(x) = 1$.
   (b) Assume that the *power rule* holds for the function $f(x) = x^k$. That is, assume $\frac{d}{dx}x^k = kx^{k-1}$.
   (c) Let $f(x) = x^{k+1}$. Notice that $f(x) = x \cdot x^k$. Using the *product rule* for dervatives, prove $f'(x) = (k+1)\,x^k$.
2. Uh oh. Our result in the last question relied on us knowing the product rule for derivatives. Prove the *product rule* for derivatives. That is, prove that if $h(x) = f(x) \cdot g(x)$ and $f(x)$ and $g(x)$ are differentiable at c, then

$$h'(c) = f(c)\,g'(c) + f'(c)\,g(c).$$

   Hint: Proceed by using the limit definition of the derivative and the Algebra of Limits Theorem for real numbers. Somewhere along the way you'll need to use the fact that g is continuous at c, and thus $\lim_{x\to c} g(x) = g(c)$.
3. Consider the following result, with which you may be familiar:

   Let $f : [a, b] \to \mathbb{R}$ be a continuous function. The maximum of f occurs either at an endpoint, at a point where f is not differentiable, or at a point $c \in [a, b]$ where $f'(c) = 0$.

   To prove the result, we will assume that the maximum does not occur at an endpoint or a point where f is not differentiable, and show that it must occur at a point where the derivative is zero.
   (a) Sketch three example functions to illustrate the result.
   (b) Before we start the proof, explain how you know that f attains a maximum.
   (c) Let f achieve its maximum at $x = c$. Assume $c \neq a$ and $c \neq b$, and that f is differentiable at $x = c$. Prove $f'(c) = 0$.

The Mean Value Theorem

4. (a) Draw a picture of a function to illustrate what the Mean Value Theorem asserts.
   (b) The Mean Value Theorem requires the function to be continuous on $[a, b]$. Draw a picture of a function that fails this condition so that the conclusion of the Mean Value Theorem does not holds true. Draw a picture of a function that fails this condition so that the conclusion of the Mean Value Theorem holds true.
   (c) The Mean Value Theorem requires the function to be differentiable on (a, b). Draw a picture of a function that fails this condition so that the conclusion of the Mean Value Theorem does not hold true. Draw a picture of a function that fails this condition so that the conclusion of the Mean Value Theorem holds true.
5. Where possible, apply the Mean Value Theorem on the following functions, using the interval $[-1, 4]$. If the hypotheses of the Mean Value Theorem hold true, find the value of c such that the theorem says must exist.
   (a) $f(x) = x^2$
   (b) $g(x) = (x+1)^a$ where $a \geq 1$
   (c) $h(x) = |x - 1|$
   (d) $k(x) = \log(2x + 4)$.
6. Use the Mean Value Theorem to prove the following statement:

   Let f be continuous on $[a, b]$ and differentiable on (a, b). If for $x \in (a, b)$ we have $f'(x) = 0$, then f is constant on $[a, b]$.

Proof of Chain Rule

7. Prove Carathéodory's Theorem:

   THEOREM:
   Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ be a function, and $c \in I$. f is differentiable at c if and only if there exists a function $\varphi : I \to \mathbb{R}$ such that φ is continuous at c and

   $$f(x) - f(c) = \varphi(x)\,(x - c)$$

   for all $x \in I$.

   In the case that f is differentiable at c, $f'(c) = \varphi(c)$.

8. Using the Carathéodory's Theorem from the previous question, prove the Chain Rule:

   Let $I, J \subseteq \mathbb{R}$ be intervals and let $f : I \to \mathbb{R}$ and $g : J \to \mathbb{R}$ be functions such that $f(I) \subseteq J$. Let $c \in I$. If f is differentiable at c and g is differentiable at $f(c)$, then $g \circ f$ is differentiable at c and $(g \circ f)'(c) = g'(f(c))\, f'(c)$.

   Here $g \circ f : I \to \mathbb{R}$ is the composition defined by $(g \circ f)(x) = g(f(x))$ for all $x \in I$.

Integration

9. Let $f : [-2, 4] \to \mathbb{R}$ be defined by $f(x) = -x + 3$. Let $P = \{-2, -1, 1, 4\}$ be a partition.
   (a) Find the upper sum $U(f, P)$.
   (b) Find the lower sum $L(f, P)$.
   (c) Find $\displaystyle\int_{-2}^{4} f(x)\, dx$ using the Fundamental Theorem of Calculus.
   (d) Do your answers satisfy the correct inequality?
   (e) Make a refinement of P and recalculate the lower and upper sums. Do all your numbers still line up?

10. Determine, without proof, which of the functions defined below are integrable on $[-1, 1]$.
    (a) $f(x) = x^7 - 5x^3$
    (b) $g(x) = \sin(e^x - 12)$
    (c) $h(x) = \begin{cases} x^2 & x \in \mathbb{Q} \\ -x^2 & x \notin \mathbb{Q} \end{cases}$
    (d) $p(x) = \begin{cases} 0 & x = 0 \\ \frac{1}{x} & x \neq 0 \end{cases}$
    (e) $q(x) = \begin{cases} x^2 + 2 & x < \frac{1}{3} \\ 9 - \log x & x \geq \dfrac{1}{3} \end{cases}$
    (f) $r(x) = \begin{cases} 17 & x = -\frac{1}{2} \\ \dfrac{1}{3x^2 + 5} & x \neq -\frac{1}{2} \end{cases}$

11. Consider the following result:

    THEOREM:
    Let $f : [a, b] \to \mathbb{R}$ be bounded. Then f is integrable on $[a, b]$ if and only if for every $\epsilon > 0$ there exists a partition P_e so that

    $$U(f, P_e) - L(f, P_e) < \epsilon$$

    Use this result to prove the following:

    Let $f : [-1, 1] \to \mathbb{R}$ be given by

    $$f(x) = \begin{cases} 0 & x \leq 0 \\ 1 & x > 0 \end{cases}$$

    By finding a partition that gives $U(f, P) - L(f, P) < \epsilon$, show that f is integrable.

    Hint: Pick a partition $P = \{-1, 0, a, 1\}$ and choose a so that you end up with $U(f, P) - L(f, P) < \epsilon$.

12. Let $f : [0,1] \to \mathbb{R}$ be given by

$$f(x) = \begin{cases} 1 & x = \frac{1}{2} \\ 0 & x \neq \frac{1}{2} \end{cases}$$

By finding a partition that gives $U(f,P) - L(f,P) < \epsilon$, show that f is integrable.

Try a similar approach to the previous question.

Improper Integrals

13. In tutorial 10B we introduce improper integrals. Use the material from that tutorial to complete the following problems.

Which of the following integrals are proper Riemann integrals, and which are improper?

(a) $\displaystyle\int_0^\infty \frac{1}{1+x^2}\,dx$

(b) $\displaystyle\int_0^{\pi/2} \tan x\,dx$

(c) $\displaystyle\int_1^2 \frac{1}{3x-1}\,dx$

(d) $\displaystyle\int_0^2 \frac{1}{3x-1}\,dx$

(e) $\displaystyle\int_{-\infty}^\infty e^{-x^4}\,dx$

(f) $\displaystyle\int_{-1}^1 \frac{\sin x}{x^2}\,dx$

(g) $\displaystyle\int_0^1 t^{1/2}e^{e^t}\,dt$

(h) $\displaystyle\int_0^1 \frac{dt}{t-\frac{1}{2}}$

(i) $\displaystyle\int_0^{100} e^{\lfloor t \rfloor}dt$ where $\lfloor t \rfloor$ is the *floor function*.

14. Evaluate the following improper integrals, or show they are divergent. Feel welcome to use tools for limits you are familiar with from previous studies.

(a) $\displaystyle\int_0^1 \frac{1}{\sqrt{x}}\,dx$

(b) $\displaystyle\int_{-1}^1 \frac{1}{x^{2/3}}\,dx$

(c) $\displaystyle\int_{-\infty}^2 \frac{1}{3-x}\,dx$

(d) $\displaystyle\int_1^\infty \frac{\log x}{x}\,dx$

(e) $\displaystyle\int_1^\infty \frac{\log x}{x^2}\,dx$

(f) $\displaystyle\int_0^1 \frac{1}{\sqrt{1-x^2}}\,dx$

(g) $\displaystyle\int_0^5 \frac{1}{\sqrt{5-x}}\,dx$

(h) $\displaystyle\int_0^\infty \frac{1}{(x+1)(x+2)}\,dx$

7. Series

Sequence Convergence and Divergence

1. For each of the following series calculate the first five partial sums and make a *guess* as to whether the series converges. It is perfectly acceptable if your guesses are incorrect.

   (a) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2}$

   (b) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n}$

   (c) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n \log n}$

   (d) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n(\log n)^2}$

   (e) $\displaystyle\sum_{n=1}^{\infty} \frac{(\log n)^2}{n^2}$

   (f) $\displaystyle\sum_{n=1}^{\infty} \frac{n^5}{2^n}$

   (g) $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{n^n}$

2. Let $a_n = \dfrac{2n}{3n+1}$.

   (a) Determine whether $\{a_n\}$ is convergent.

   (b) Is $\displaystyle\sum_{n=1}^{\infty} a_n$ is convergent?

3. The following series can all be written as geometric series, perhaps after a little algebra. Determine which of the series below converge. For those that do, determine to which value.

   (a) $\displaystyle\sum_{n=0}^{\infty} \left(\frac{4}{5}\right)^n$

   (b) $\displaystyle\sum_{n=0}^{\infty} \frac{2^{n+2}}{6^{2n+1}}$

   (c) $\displaystyle\sum_{n=2}^{\infty} \left(-\frac{1}{3}\right)^n$

4. Since series are sequences in disguise, we can use comparison techniques to determine their convergence. In Tutorial 11 we introduce the following theorem:

   Theorem *Assume* $0 \le a_n \le b_n$ *for all* $n \in \mathbb{N}$.

   - If $\displaystyle\sum_{n=0}^{\infty} b_n$ converges, then $\displaystyle\sum_{n=0}^{\infty} a_n$ converges
   - If $\displaystyle\sum_{n=0}^{\infty} a_n$ diverges, then $\displaystyle\sum_{n=0}^{\infty} b_n$ diverges

   Use comparisons, determine whether the series converges or diverges.

   (a) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2+1}$

   (b) $\displaystyle\sum_{n=1}^{\infty} \frac{2n}{n^2+1}$

   (c) $\displaystyle\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+2}$

   (d) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n\,4^n}$

   (e) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{1+3^n}$

   (f) $\displaystyle\sum_{n=1}^{\infty} \frac{\sin^2(nx)}{n^2}$

   (g) $\displaystyle\sum_{n=2}^{\infty} \frac{3}{\sqrt[3]{n^2-2}}$

   Extra: in part 4f, are there any special values of x for which you can evaluate the series exactly?

5. Prove the theorem from the previous question.

6. Use the limit ratio test to determine the convergence or divergence of the series.

   (a) $\displaystyle\sum_{n=1}^{\infty} \frac{n^2}{e^n}$

   (b) $\displaystyle\sum_{n=1}^{\infty} \frac{n^3}{2^n}$

   (c) $\displaystyle\sum_{n=1}^{\infty} \frac{2^n}{n+1}$

7. In our discussion about the integral test, we computed our integral from the lower bound of $a = 1$. It turns out that removing any number of terms from the front of a series does not affect whether the series converges or diverges. And so we have the following result

   Theorem *Let (a_n) be a sequence of positive terms. Suppose that $a_n = f(n)$, where f is a integrable, positive, decreasing function of x for all $x \geq N$, $n \in \mathbb{N}$. Then the series $\sum_{n=N}^{\infty} a_n$ and the integral $\int_N^{\infty} f(x)\,dx$ either both converge or both diverge.*

   (a) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n \log n}$ (b) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n \log n \log(\log n)}$ (c) $\displaystyle\sum_{n=1}^{\infty} \frac{n}{n^2+2}$ (d) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2+9}$

8. Determine whether the following alternating series converge or diverge:

   (a) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}\,2n}{4n^2-3}$ (b) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}\,2n}{4n-3}$ (c) $\displaystyle\sum_{n=1}^{\infty} (-1)^n$ (d) $\displaystyle\sum_{n=1}^{\infty} (-1)^n \frac{1+3^n}{1+4^n}$

9. Is it true that the convergence of $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ implies the convergence of $\sum_{n=1}^{\infty} (a_n + b_n)$? If it is true, prove it. If it is not true, find a counterexample.

10. Is it true that the convergence of $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ implies the convergence of $\sum_{n=1}^{\infty} a_n b_n$? If it is true, prove it. If it is not true, find a counterexample.

Power Series

11. Find the interval of convergence and radius of convergence for each of the series below.

    (a) $\displaystyle\sum_{n=1}^{\infty} \frac{x^n}{n^2}$ (c) $\displaystyle\sum_{n=0}^{\infty} \frac{x^{2n+1}}{n!}$ (e) $\displaystyle\sum_{n=0}^{\infty} \frac{x^n}{2n+1}$

    (b) $\displaystyle\sum_{n=0}^{\infty} \frac{n^2}{2^n} x^n$ (d) $\displaystyle\sum_{n=1}^{\infty} \frac{(-3)^n}{n} x^{n+1}$

Taylor Series and Power Series

12. Find the Taylor series at 0 and interval of convergence each given function

    (a) $\sin(2x)$ (c) $\dfrac{1}{1+x}$ (e) $\log(2x+1)$

    (b) $\cos x$ (d) $\sinh x$ (f) $(1+x)^{-2}$

13. Find the Taylor series for each function f at b.

    (a) $f(x) = \sin x; \quad b = \frac{1}{4}\pi$ (c) $f(x) = e^x; \quad b = -3$

    (b) $f(x) = 1/x; \quad b = 2$

14. Consider the function $f : \mathbb{R}\backslash\{-1\} \to \mathbb{R}$ given by

$$f(x) = \frac{1}{1+x}$$

    (a) Find the nth order Taylor polynomial at 0 and an expression for the error this approximation would give.

    (b) For $0 < x < 1$ show that the error approaches 0 as $n \to \infty$.

    (c) Prove $f(x)$ is equal to its Taylor series at $x = 0$ for $0 < x < 1$.

15. Estimate $\displaystyle\int_0^1 \left(\frac{\sin t}{t}\right) dt$ using Taylor polynomials for $\sin t$ at 0 of

    (a) degree 3 (b) degree 5.

    Use Taylor's theorem to estimate the error in these approximations.

16. Here we give an example of function f which has a convergent Taylor series, but with the Taylor series *not* equal to the function! Define

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}} & x \neq 0\,, \\ 0 & x = 0\,. \end{cases}$$

    (a) Show that f is continuous at 0.

    (b) Show that for any $m \in \mathbb{N}$

$$\lim_{x\to 0} \frac{f(x)}{x^m} = 0\,.$$

    (c) Show that

$$f'(x) = \begin{cases} \frac{2}{x^3}\, e^{-\frac{1}{x^2}} & x \neq 0\,, \\ 0 & x = 0\,. \end{cases}$$

    (d) Use induction to show that for every $n \in \mathbb{N}$ there is a rational function R such that

$$f^{(n)}(x) = \begin{cases} R(x)\, e^{-\frac{1}{x^2}} & x \neq 0\,, \\ 0 & x = 0\,. \end{cases}$$

    (e) What is the Taylor series for $f(x)$ at 0? How good an approximation is it?

1. Proof and Mathematical Logic - Selected Answers

Logic and Notation

1. (a) "A monkey is happy" $\implies$ "The monkey is eating a banana."
   (b) "You don't step on the tail of a snake" $\implies$ "The snake will not bite you."
   (c) "A donkey sees a mule" $\implies$ "The donkey laughs at him."
   (d) Wealth $\implies$ Happiness
   (e) Happiness $\implies$ Wealth

2. (a) False: Neptune is a planet, not a moon.
   (b) True: Jupiter *is* a planet.
   (c) False: both parts are false.
   (d) True: at least one of the parts is true (in this case, both are true).
   (e) True
   (f) True: Napolean was born in Corsica, France. The second part happens to be true, although it doesn't matter in this case.
   (g) False: Dodos *are* unfortunately extinct (hunted to extinction by man), but pigs cannot fly.
   (h) True

3. Let $P =$ the set of prime numbers.
   (a) $(\sim (6 \in P)) \vee (\sim (11 \in P))$
   (b) $(10^2 = 50) \wedge (5^3 = 12)$
   (c) $(7 \in \mathbb{Z}) \implies (\sim (6 \in \mathbb{Z}))$
   (d) $((2 \in P) \wedge (5 \in P)) \implies (\sim (2 \times 5 \in P))$

4. (a) The truth table for $(p \wedge q) \vee (\sim p \wedge \sim q)$ is:

p	q	$(p \wedge q) \vee (\sim p \wedge \sim q)$
T	T	T
T	F	F
F	T	F
F	F	T

This one is logically equivalent to the biconditional connective.

   (b) The truth table for $[\sim q \wedge (p \implies q)] \implies \sim p$ is:

p	q	$[\sim q \wedge (p \implies q)] \implies \sim p$
T	T	T
T	F	T
F	T	T
F	F	T

This one is a tautology.

(c) The truth table for $[(p \vee q) \wedge r] \implies (p \wedge r)$ is:

p	q	r	$[(p \vee q) \wedge r] \implies (p \wedge r)$
T	T	T	T
T	T	F	T
T	F	T	T
T	F	F	T
F	T	T	F
F	T	F	T
F	F	T	T
F	F	F	T

5. (a) $(\forall r \in \mathbb{Q})\ \ r > 6$ This is false.

   (b) $(\exists x \in \mathbb{R})\ \ x^2 + 3x - 7 = 0$ This is true.

   (c) $(\exists n \in \mathbb{N})\ \ n^3 = 8$ This is true.

6. There are of course many answers.

   (a) "Adding zero to any rational number doesn't change the number."

   (b) "The square of any real number is greater than one."

7. The second statement is the true one.

8. (a) $(\exists x \in \mathbb{R})\ \ x^2 \neq 10$

   (b) $(\forall y \in \mathbb{N})\ \ y \geq 0$

   (c) $(\forall a \in \mathbb{N})\ [(\exists x \in \mathbb{R})\ \ ax \neq 4]$

   (d) $(\exists y \in \mathbb{Q})\ [(\forall x \in \mathbb{R})\ \ x/y \neq 30]$

9. (a) The truth table for $p \wedge q \iff q \wedge p$ is:

p	q	$p \wedge q$	$q \wedge p$	$p \wedge q \iff q \wedge p$
T	T	T	T	T
T	F	F	F	T
F	T	F	F	T
F	F	F	F	T

As the biconditional is a tautology, the two statements $p \wedge q$ and $q \wedge p$ are logically equivalent.

(b) The truth table for $\sim (p \implies q) \equiv (p \wedge \sim q)$ is:

p	q	$\sim (p \implies q)$	$p \wedge \sim q$	$\sim (p \implies q) \equiv (p \wedge \sim q)$
T	T	F	F	T
T	F	T	T	T
F	T	F	F	T
F	F	F	F	T

As the biconditional is a tautology, the two statements $\sim (p \implies q)$ and $p \wedge \sim q$ are logically equivalent.

Both (i) and (ii) are tautologies.

Counter Examples

10. (a) Consider $3 \times 4 = 12$ which is even, but 3 is odd, so the statement is false.

    (b) Put $x = 1$ and $y = -1$. Then $x^2 = y^2$, but $x \neq y$.

    (c) Put $a = 11$, so $a = 7 + 4$ and the remainder is 4. But $5a = 55 = 7 \times 7 + 6$ and the remainder is $6 \neq 4$.

Direct Proofs

11. These proofs will all follow the same basic patter. Identify the premise and express it in mathematical notation; identify the conclusion and express it in mathematical notation; connect the two with a reasoned argument. For example in (a):

    Claim: The product of an even integer with an odd integer is even.

    Proof:

    1. Let x be an even integer. (premise)
    2. There exists $k \in \mathbb{Z}$ such that $x = 2k$ (1, defn of even)
    3. Let y be an odd integer. (premise)
    4. There exists $\ell \in \mathbb{Z}$ such that $y = 2\ell + 1$ (3, defn of odd)
    5. $xy = (2k)(2\ell + 1)$ (2,4)
    6. $xy = 2(2k\ell + k)$ (algebra)
    7. xy is even (defn of even, 6)
    8. If x is even and y is odd, then xy is even. (1,3,7)

Domains

12. The sets that go with each equation are:

    (a) $\mathbb{R}$
    (b) $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$
    (c) $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$
    (d) none
    (e) $\mathbb{Q}, \mathbb{R}$
    (f) none

Contrapositive

13. Here's the proof of Part (a).

    Claim: If n^4 is even, then n is even.

    Proof: We proceed by proving the contrapositive. Assume n is odd. Since n is odd, there exists $k \in \mathbb{Z}$ such that $n = 2k + 1$. We compute

    $$(2k + 1)^4 = 16k^4 + 32k^3 + 24k^2 + 8k + 1 = 2(8k^4 + 16k^3 + 12k^2 + 4k) + 1$$

    Therefore, n^4 is odd.

    If n is odd, then n^4 is odd. And so, if n^4 is even, it must be that n is even.

Proof by Contradiction

14.

15.

16. There are 3 paintings which satisfy the special rule.

More Proofs

17. Here are some hints:

    (a) Show the result for the cases $n = 3k + 1$ and $n = 3k + 2$. See the lecture solution of the square of an odd integer is odd and use the same idea for each case separately.

    (b) There are two cases here, m and n both odd or both even.

Irrational numbers

18. (a) Mimic the proof of the irrationality of $\sqrt{2}$.

    (b) Assume it is rational, so that $\sqrt{15} + \sqrt{5} = p/q$. Simplify and find an expression for $\sqrt{3}$ as a rational number. Finally, use Part (a).

    (c) Proceed as in Part (b).

Mathematical Induction

19. Mimic the proofs from lectures and tutorials.

    The equations, rewritten using summation notation, are:

    (a) $\sum_{k=1}^{n}(5k-3) = \frac{1}{2}n(5n-1)$

    (b) $\sum_{k=1}^{n} k2^{k-1} = 1 + (n-1)\,2^n$

    (c) $\sum_{k=1}^{n} \dfrac{1}{k(k+1)} = \dfrac{n}{n+1}$

    (d) $\dfrac{a^n - b^n}{a - b} = \sum_{k=1}^{n} a^{n-k}b^{k-1}$

20. Here is a proof of Part (a).

    We proceed by induction. Let $p(n)$ be the statement "there exists $m \in \mathbb{Z}$ such that $n^3 - n + 3 = 3m$". The statement $p(0)$ is true as $0^3 - 0 + 3 = 3 = 3(1)$.

    Let k be an integer. Assume that $p(k)$ is true. That is, assume that there exists $m \in \mathbb{N}$ such that $n^3 - n + 3 = 3m$. Consider the expression $(k+1)^3 - (k+1) + 3$. Algebraically we find that

    $$\begin{aligned}(k+1)^3 - (k+1) + 3 &= k^3 + 3k^2 + 2k + 3 \\ &= (k^3 - k + 3) + 3(k^2 + 3k) \\ &= 3p + 3(k^2 + k) && (p(k) \text{ is true}) \\ &= 3(p + k^2 + k)\end{aligned}$$

    Therefore 3 divides $(k+1)^3 - (k+1) + 3$. Therefore $p(k+1)$ is true.

    The result now follows by the Principle of Mathematical Induction.

    The other parts of this question are similar, except (d). Recall that when we talked about factors of polynomials, we mean something different to factors of integers.

21.

22. Here is the inductive step for (a).

    Assume that there exists $k \in \mathbb{N}$ such that $1 + 2k \leq 3^k$.

    Claim: $1 + 2(k+1) \leq 3^{k+1}$

Proof: First note that $1 + 2(k + 1) = 3 + 2k$

$$\begin{aligned} 3^{k+1} &= 3 \cdot 3^k \\ &\geq 3(1 + 2k) \quad \text{from inductive premise} \\ &= 3 + 6k \\ &= 3 + 2k + 4k \\ &\geq 3 + 2k \quad \text{for } k \geq 0 \\ &= 1 + 2(k + 1). \end{aligned}$$

Therefore $1 + 2(k + 1) \leq 3^{k+1}$.

23.

2. Set Theory - Selected Answers

Sets

For questions 1 to 3, let $A = \{1, 2, 3, 4\}$, $B = \{1, 3, 5, 7\}$ and $C = \{2, 3\}$.

1. (a) $\{1, 2, 3, 4, 5, 7\}$
   (b) $\{1, 3\}$
   (c) $\{1, 2, 3, 4\} = A$
   (d) $\{2, 3\} = C$
   (e) $\{1, 2, 3, 4, 5, 7\} = A \cup B$
   (f) $\{3\}$

2. (a) $\{(1, 2), (1, 3), (3, 2), (3, 3), (5, 2), (5, 3), (7, 2), (7, 3)\}$ (b) $\{\}$

3. (a) False: 3 is not a set, so it can't be a subset of anything. However, we could say 3 is an element of B ($3 \in B$).
   (b) True: the empty set is a subset of all sets.
   (c) True
   (d) False: 2 is in C, but not in B.
   (e) False: B contains numbers which aren't in C.
   (f) False: A does not contain sets of numbers, so no set is an element of A. However, we could say that $C \subseteq A$.
   (g) True
   (h) False: The number 1 is in B, and there is no number in C you can subtract from it that gives a result that is non-negative.

4. General hints:
   - To prove that $A = B$ you usually need to prove $A \subseteq B$ and $B \subseteq A$.
   - To prove that $A \subseteq B$, start with $x \in A$ as a premise and try to prove that $x \in B$. Use the definition of subset to conclude $A \subseteq B$.
   - One way to prove set theorems is to rewrite the theorem in logic form and then prove the logic form eg. $A \subseteq B$ is true if and only if $\forall x \in \mathcal{U} \quad [(x \in A) \implies (x \in B)]$ is true.

   Hints for specific subquestions:
   (a) Use the axiom $x \in B \vee \sim x \in B$
   (c) You may need to use the logical equivalence $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

5. (a) The length is 1.
   (b) $C_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$
   (c) $C_2 = [0, \frac{1}{9}] \cup [\frac{2}{9}, \frac{1}{3}] \cup [\frac{2}{3}, \frac{7}{9}] \cup [\frac{8}{9}, 1]$

Supremum and Infimum

6. (a) $\sup S = 3$ and $\inf S = -3$. Both are in S.
   (b) $\sup S = 5$ and $\inf S = -1$. Neither is in S.
   (c) $\sup S = 2$ and $\inf S = -3$. Neither is in S.
   (d) $\sup S = 0$ and $\inf S = -1$. Neither is in S.
   (e) No supremum or infinum.
   (f) $\sup S = \sqrt{7}$ and $\inf S = -\sqrt{7}$. Neither is in S.

7. Here is a proof of the first half of part (a).
   Assume S is bounded above. Therefore there exists $\beta \in \mathbb{R}$ so that for every $x \in S$ we have $x \leq \beta$. Therefore $x + c \leq \beta + c$ for every $x \in S$. Therefore $\beta + c$ is an upper bound for $c + S$. Therefore $c + S$ is bounded above.

8.

Inequalities

Recall the following:

The Triangle Inequality: For all $a, b \in \mathbb{R}$ we have $|a + b| \leq |a| + |b|$.

9.
   (a) True
   (b) True
   (c) True
   (d) True only if $c > 0$. What if $c = 0$ or $c < 0$?
   (e) Sometimes true (Hint: $a < 0 < b$ vs $0 < a < b$)
   (f) Sometimes true
   (g) True only if $a > 0$.
   (h) False. What should it be?

10. (a) $x \in [-\frac{5}{2}, \frac{3}{2}]$
    (b) $x \in (-\infty, -7] \cup [3, \infty)$
    (c) $x \in (2, \infty)$
    (d) $x \in (-2, 5)$
    (e) $x \in (-1, 0)$

11. There are many correct answers here, but these answers are ones you might reasonably get:
    (a) $\frac{3}{2}$
    (b) $\frac{37}{9}$

12.

13.

14. (a) Consider $a = 0$. The right hand side will be positive. Repeat for $b = 0$.
    (b) Remember to check the second derivative to make sure it's is a minimum, and not a maximum.
    (c) Consider $x = b^{\frac{1}{p-1}}$ and $q = p/(p-1)$ and use a bit of algebra to simplify.
    (d) Replace x with a and, with a small amount of rearranging, voila!

3. The Real Numbers - Selected Answers

Proofs with the Real Number Axioms

1. (a) i. Let $x, y, z \in \mathbb{R}$ so that $x + z = y + z$ (premise)

   ii. There exists $w \in \mathbb{R}$ so that $z + w = 0$ (A5)

   iii. $(x + z) + w = (y + z) + w$ (OA)

   iv. $x + (z + w) = y + (z + w)$ (A3)

   v. $x + 0 = y + 0$ (A5)

   vi. $x = y$ (A4)

   vii. Therefore, if $x + z = y + z$, then $x = y$.

   (b) Hint: First show that $0 + 0 = 0$ and then multiply both sides by x.

   If you pay attention, you may be able to use the result of (a) to save time.

   (c) Hint: Start with $x + (-x) = 0$ and add $-(-x)$ to both sides.

   (d) Hint: Multiply both sides by $1/b$. What do we need to assume to justify this?

   (e) Hint: Show that both $y - x$ and $-z$ are positive and then multiply them using (OM).

   (f) Hint: Otherwise $1/x < 0$. (Why?) Multiply both sides by x.

Properties of Number Sets

2. (a) The integers have additive inverses, but the natural numbers do not. In particular, this lets us define *subtraction* of integers.

   (b) The rational numbers (besides zero) have multiplicative inverses, but the integers do not (except 1 and -1). In particular, this lets us define *division* of rational numbers.

   (c) The real numbers satisfy the Completeness Axiom, but the rational numbers do not.

The Archimedean Principle

3. Use v2 of the Archimedean Principle with $y = 1$ and $x = \epsilon$

Existence of the square root of 2

4. (a) Let $\beta = 2$. We want to show that for every $x \in A$, $x \leq 2$. Equivalently, we can show that for each $x \in \mathbb{R}$ if $x > 2$ then $x \notin A$. Let $x \in \mathbb{R}$ such that $x > 2$. Then using Theorem 3.8, (M2), and (M3), $x^2 < 2x < 2^2 < 1 \cdot 2 = 2$. Hence $x \notin A$. Therefore, for each $x \in \mathbb{R}$ if $x > 2$ then $x \notin A$. Therefore, 2 is an upper bound of A.

   (b) Hint: Use the Completeness Axiom. What conditions do you need check to apply the Completeness Axiom?

   (c) (i) As scrap work, we have that

$$(\alpha + 1/n)^2 = \alpha^2 + \frac{2\alpha}{n} + \frac{1}{n^2} < 2.$$

   Since $\frac{1}{n^2} < \frac{1}{n}$, we could require

$$\alpha^2 + \frac{2\alpha}{n} + \frac{1}{n} < 2 \quad \Rightarrow \quad \frac{2\alpha + 1}{2 - \alpha^2} < n.$$

Proof. By the Archimedean Principle, there exists $n \in \mathbb{N}^+$ such that $\frac{2\alpha+1}{2-\alpha^2} < n$. Hence

$$\begin{aligned}
\frac{2\alpha+1}{2-\alpha^2} &< n \\
\frac{2\alpha+1}{n} &< 2-\alpha^2 \\
\alpha^2 + \frac{2\alpha}{n} + \frac{1}{n} &< 2 \\
\alpha^2 + \frac{2\alpha}{n} + \frac{1}{n^2} &< 2 \qquad \text{(since } n \geq 1\text{)} \\
(\alpha + 1/n)^2 &< 2.
\end{aligned}$$

(ii) By Part (c)(i), $(\alpha + 1/n)^2 < 2$. Hence $\alpha + 1/n \in A$. Since $\alpha + 1/n \in A$ and $\alpha + 1/n > \alpha$, α cannot be an upper bound of A.

(d) Similar to Part (c).

(e) Since $1 \in A$ and $\alpha = \sup A$, $0 < 1 \leq \alpha$. By (O1), either $\alpha^2 < 2$, $\alpha^2 > 2$, or $\alpha^2 = 2$. Hence by Parts (c) and (d), $\alpha^2 = 2$.

4. Functions and Sequences - Selected Answers

ϵ–M Definition of Convergence

1. Hints:

   (a) Choose $M > \left\lceil \frac{1}{\epsilon} \right\rceil - 2$

   (b) Choose $M > \left\lceil \frac{1}{\sqrt{\epsilon}} \right\rceil - 4$

   (c) Choose $M > \left\lceil \frac{\sqrt{(2/\epsilon)-4}}{2} \right\rceil$

   These answers are not unique. We do not need to choose the smallest possible value of M. For example, in (a) one could choose $M > \left\lceil \frac{1}{\epsilon} \right\rceil$ as $\left\lceil \frac{1}{\epsilon} \right\rceil > \left\lceil \frac{1}{\epsilon} \right\rceil - 2$

2. Limits:

   (a) 0 (b) 2 (c) $\frac{3}{2}$ (d) $\frac{1}{2}$

3. Proceed as in the proof of part i of the Algebra of Limits Theorem. Notice $2u_n = u_n + u_n$.
4. Consider the sequence $(1, 1, 1, 1, \dots)$.
5. Given ϵ we can choose M_a and M_c so that

$$|a_{n_a} - L| < \epsilon \text{ whenever } n_a > M_a$$

   and

$$|c_{n_c} - L| < \epsilon \text{ whenever } n_c > M_c$$

   Choose $M = \max\{M_a, M_c\}$

Subsequences

6. (a) The sequence is unbounded. For any $r \in \mathbb{R}$ we can find $n \in \mathbb{N}^+$ so that $f_n > r$. (In particular, choose n so that $n > r^2$)

   (b) There are two convergent subsequences.

7. (a) $-1,\ 1;$ divergent. (b) $-1,\ 0,\ 1;$ divergent. (c) $0;$ convergent.

8. Hint: for every $\epsilon > 0$, there exists $\epsilon' > 0$ so that $\epsilon = |k|\epsilon'$.
9. Let $\epsilon > 0$. Suppose K_e and K_o are positive integers such that $|a_{2k} - L| < \epsilon$ for all $k > K_e$ and $|a_{2k-1} - L| < \epsilon$ for all $k > K_o$. Use these to find M such that etc.

Monotone Sequences

10. (a) monotone increasing

    (b) neither

    (c) montone increasing

    (d) both

    (e) monotone decreasing

11. (a) and (c) do not converge by virtue of being unbounded. (d) and (e) converge by virtue of being bounded.
12. If $a_n \leq a_{n+1}$ and $a_n \geq a_{n+1}$ for all $n \in \mathbb{N}^+$, then $a_n = a_{n+1}$ for all $n \in \mathbb{N}^+$. Therefore every term of (a_n) is equal to a_1:

$$(a_1, a_1, \dots)$$

    Therefore $a_n \to a_1$.

Cauchy Sequences

13. Choose M so that $\frac{1}{M} < \frac{3\epsilon}{2}$

14. One possible M is from solving $|f_n - f_m| < \dfrac{7m+7n}{mn} = \dfrac{7}{n} + \dfrac{7}{m} < \dfrac{\epsilon}{2} + \dfrac{\epsilon}{2}$. Then pick $M > \dfrac{14}{\epsilon}$.

Contractive Sequences

15. (a) For the inductive step, note that $1/2 \leq x \leq 1$ implies $1 < 1 + x \leq 2$.

    (b) From the previous part, conclude that $1/2 \leq 1/(1+f_n) \leq 2/3$ for all n. A bit of algebra shows that $|f_{n+2} - f_{n+1}| = \frac{|f_{n+1}-f_n|}{(1+f_{n+1})(1+f_n)}$. Combine these two to obtain the bound.

    (c) The sequence is contractive, and therefore convergent. Let f be the limit. Applying the Algebra of Limits Theorem, we see that $f = 1/(1+f)$, or $f^2 + f - 1 = 0$. This quadratic equation has one positive root $f = (\sqrt{5}-1)/2$. (This is the reciprocal of the Golden Ratio.)

16. Choose some x_1 with $0 < x_1 < 1$, and inductively define a sequence by $x_{n+1} = \frac{1}{6}(x_n^3 + 3)$. Proceeding as in the example given in lectures, you can show that this sequence is contractive with $c = 1/2$, and $0 < x_n < 1$ for all n. If L is its limit, then $0 \leq L \leq 1$ and the Algebra of Limits Theorem shows that $L = \frac{1}{6}(L^3 + 3)$, and thus $L^3 - 6L + 3 = 0$.

Existence

17. (a) If (f_n) is bounded then there exists $C \in \mathbb{R}$ so that $|f_n| \leq C$ for all $n \in \mathbb{N}^+$. Therefore the terms of any subsequence (f_{n_k}) must also be bounded by C. Thus no example exists.

    (b) $(1, 0, 2, 0, 3, 0, 4, \dots)$

    (c) $(1, 2, 3, \dots)$

    (d) $(-1, 1, -1, \dots)$

    (e) Does not exist. A bounded monotone sequence must converge. And so must be Cauchy

    (f) Does not exist. A Cauchy sequence must converge. All convergent sequences are bounded.

    (g) $\left(\frac{(-1)^n}{n}\right)$

    (h) Does not exist. A Cauchy sequence must converge. All subsequences of a convergent sequence must converge. All convergent sequences are bounded.

    (i) $(1, 0, 2, 0, 3, 0, 4, \dots)$

    (j) A divergent monotone subsequence is not bounded. See (a)

    (k) $(-1, 1, -1, \dots)$

    (l) $(100, -100, 0, 0, \dots)$

    (m) Does not exist. In the proof that a bounded increasing monotone converges, one shows that it converges to $\sup\{f_n \mid n \in \mathbb{N}^+\}$

5. Limits and Continuity - Selected Answers

Limit Points

1. (a) 0
   (b) $\left\{ \frac{1}{n} \middle| n \in \mathbb{N} \right\} \cup \{0\}$
   (c) $1, -1$
   (d) None
   (e) $\mathbb{R}$

2. (a) Let $E \subseteq \mathbb{R}$ and let $a \in \mathbb{R}$. Let (x_n) be a sequence such that $x_n \in E$ and $x_n \neq a$ for all $n \in \mathbb{N}^+$ and $x_n \to a$ as $n \to \infty$.

   Let $\epsilon > 0$. Since $x_n \to a$

   $$\text{there exists } M \in \mathbb{N}^+ \text{ such that if } n \in \mathbb{N}^+ \text{ with } n > M \text{ then } |x_n - a| < \epsilon.$$

   Therefore, $x_n \in E$, $x_n \neq a$, and $|x_n - a| < \epsilon$. That is, there exists a point $x \in E$ (namely $x = x_n$) such that $x \neq a$ and $|x - a| < \epsilon$. Therefore, a is a limit point of E.

   (b) Next let's show the "only if" part. Suppose that a is a limit point of E.

   (i) Since a is a limit point of E

   $$\text{for every } \delta > 0 \text{ there exists } x \in E \text{ such that } x \neq a \text{ and } |x - a| < \delta.$$

   Let $\delta = 1/n$. Then there exists $x_n \in E$ such that $x_n \neq a$ and $|x_n - a| < 1/n$.

   (ii) Hint: Use the Archimedean Principle.

ϵ-δ Limits

3. Possible choices for δ:

   (a) $\dfrac{\epsilon}{5}$
   (b) $\dfrac{\epsilon}{5}$
   (c) $\dfrac{\epsilon}{3}$
   (d) $\min\{1, 2\epsilon\}$

   Proof for (a): Let $\epsilon > 0$. Choose $\delta = \dfrac{\epsilon}{5}$.

   Assume $0 < |x - 3| < \delta$.

   Then
   $$\begin{aligned} |5x - 15| &= 5|x - 3| \\ &< 5 \cdot \frac{\epsilon}{5} \\ &= \epsilon. \end{aligned}$$

   Therefore $0 < |x - 3| < \delta$ implies $|5x - 15| < \epsilon$.

Limits to Infinity

4. Possible choices for M are:

   (a) $M = 1$
   (b) $M = \epsilon^{-1/p}$

5. Use $M = \frac{1}{\epsilon^2}$.
6. When $x > r^2$ we have $\sqrt{x} > r$.
7.

Weird Functions

8. Hint: Proceed by contradiction and consider $\epsilon = 1/2$.

9. Hint: $0 < |x - 0| < \epsilon$ is true for every $x \in \mathbb{Q}$. And so one need only consider $x \notin \mathbb{Q}$.

10. (i) We claim that $\lim_{x \to a} f(x) = 0$ for all $a \in \mathbb{R}$.

    Let $a \in \mathbb{R}$. Let $\epsilon > 0$. The challenge is to find $\delta > 0$ such that for each $x \in \mathbb{R}$

    $$0 < |x - a| < \delta \quad \Longrightarrow \quad |f(x) - 0| < \epsilon.$$

    By the Archimedean Principle, there exists $n \in \mathbb{N}^+$ such that $1/n < \epsilon$. Given an integer q such that $1 \leq q \leq n$, let

    $$\delta_q = \inf \left\{ \left| \frac{p}{q} - a \right| : p \in \mathbb{Z} \text{ and } \frac{p}{q} \neq a \right\}.$$

    That way for each $p \in \mathbb{Z}$, either $p/q = a$ of $|p/q - a| \geq \delta_q$. To see that $0 < \delta_q < \infty$, by again applying the Archimedean Principle there exists $k \in \mathbb{Z}$ such that $k \leq qa < k + 1$; that is, $k/q \leq a < (k + 1)/q$. It follows that in the case $a = k/q$, $\delta_q = 1/q$. In the case $a \neq k/q$,

    $$\delta_q = \min \left\{ a - \frac{k}{q}, \frac{k+1}{q} - a \right\} > 0.$$

    Let $\delta = \min\{\delta_1, \delta_2, \ldots, \delta_n\} > 0$. Let $x \in \mathbb{R}$ such that $0 < |x - a| < \delta$. If $x \notin \mathbb{Q}$, then

    $$|f(x) - 0| = |0 - 0| = 0 < \epsilon.$$

    Suppose that $x \in \mathbb{Q}$ and $x = p/q$, where p, q are relatively prime integers and $q > 0$. If $q \leq n$, then either $x = a$ or $|x - a| \geq \delta_q \geq \delta$, contrary to assumption. Hence we must have $q > n$. (This includes the case $x = 0$, as $0 = 0/1$ with $q = 1$.) Thus

    $$|f(x) - 0| = |1/q - 0| = 1/q < 1/n < \epsilon.$$

    Therefore, if $0 < |x - a| < \delta$ then $|f(x) - 0| < \epsilon$. That is, $\lim_{x \to a} f(x) = 0$.

    (ii) Let $a \in \mathbb{R}$. Since the domain of f is $\mathbb{R}$, f is continuous at $a \iff \lim_{x \to a} f(x) = f(a)$. By Part (a), $\lim_{x \to a} f(x) = 0$. Hence f is continuous at $a \iff f(a) = 0 \iff a \notin \mathbb{Q}$.

Continuity

11. (a) Let $\epsilon > 0$. Since f and g are continuous at a, there exists $\delta_f, \delta_g > 0$ so that

    - $|f(x) - f(a)| < \epsilon$ whenever $x \in (a - \delta_f, a + \delta_f)$; and
    - $|g(x) - f(a)| < \epsilon$ whenever $x \in (a - \delta_g, a + \delta_g)$.

    Let $\delta = \min\{\delta_f, \delta_g\}$. Use the Triangle Inequality to prove:

    $$|f(x) - g(x) - f(a) - g(a)| < \epsilon$$

    whenever $x \in (a - \delta, a + \delta)$.

    (b)

12. Is the function f is continuous on the interval $[-1, 1]$?

    (a) Yes (b) No (c) No

13. The converse is not true. Consider the irrational indicator function:

    $$f(x) = \begin{cases} 1 & x \notin \mathbb{Q} \\ -1 & x \in \mathbb{Q} \end{cases}$$

The Intermediate Value Theorem

14. (a) **Claim:** There exists $c \in \mathbb{R}$ such that $f(c) = 100$.

    Proof: Since f is a polynomial, f is continuous for all $x \in \mathbb{R}$.

    Notice $f(0) = -9 < 100$ and $f(10) = 561 > 100$.

    Therefore, by the Intermediate Value Theorem, there exists $c \in (0, 10)$ such that $f(c) = 100$.

    (b) Find $f(0)$ and $f(1)$. Notice they have opposite signs

    (c) Let $f(x) = x + \sin x - 1$. Compare the sign of this function at $x = 0$ and $x = \pi/6$.

    (d) Let $g(x) = x - f(x)$. We have $g(0) = 0 - f(x) \leq 0$ and $g(1) = 1 - f(x) \geq 0$. Applying the Intermediate Value Theorem implies there exists $c \in [0, 1]$ so that $g(c) = 0$. Therefore $f(c) = c$

    (e) Proceed as in the previous question. Consider $g(x) = f(x + 1) + f(x)$.

6. Differentiation and Integration - Selected Answers

The Definition of the Derivative

1.

2.

$$\begin{aligned}
\lim_{x\to c} \frac{f(x)\,g(x) - f(c)\,g(c)}{x-c} &= \lim_{x\to c} \frac{f(x)\,g(x) - f(c)\,g(x) + f(c)\,g(x) - f(c)\,g(c)}{x-c} \\
&= \lim_{x\to c} \left(\frac{f(x)\,g(x) - f(c)\,g(x)}{x-c} + \frac{f(c)\,g(x) - f(c)\,g(c)}{x-c} \right) \\
&= \lim_{x\to c} \frac{f(x)\,g(x) - f(c)\,g(x)}{x-c} + \lim_{x\to c} \frac{f(c)\,g(x) - f(c)\,g(c)}{x-c}
\end{aligned}$$

Since g is differentiable at c, it follows that g is continuous at c. Therefore $\lim_{x\to c} g(x) = g(c)$. And so

$$\begin{aligned}
\lim_{x\to c} \frac{f(x)\,g(x) - f(c)\,g(c)}{x-c} &= \lim_{x\to c} g(x)\,\frac{f(x)-f(c)}{x-c} + \lim_{x\to c} f(c)\,\frac{g(x)-g(c)}{x-c} \\
&= \left(\lim_{x\to c} g(x)\right)\left(\lim_{x\to c} \frac{f(x)-f(c)}{x-c}\right) + f(c) \lim_{x\to c} \frac{g(x)-g(c)}{x-c} \\
&= g(c)\left(\lim_{x\to c} \frac{f(x)-f(c)}{x-c}\right) + f(c) \lim_{x\to c} \frac{g(x)-g(c)}{x-c} \\
&= g(c)\,f'(c) + f(c)\,g'(c)
\end{aligned}$$

3. The function f attains a maximum by the Extreme Value Theorem. Prove

$$\lim_{x\to c^-} \frac{f(x)-f(c)}{x-c} \geq 0 \quad \text{and} \quad \lim_{x\to c^+} \frac{f(x)-f(c)}{x-c} \leq 0$$

The Mean Value Theorem

4.

5. By the Mean Value Theorem, there exists $c \in (-1, 4)$ so that:
   (a) $f'(c) = 3$
   (b) $g'(c) = 5^{a-1}$
   (c) The Mean Value Theorem does not apply to the function h because it is not differentiable at $x = 1$.
   (d) $k'(c) = \frac{1}{5}\log 6$.

6. Let $x \in (a, b]$ and apply the Mean Value Theorem to f on the interval $[a, x]$.

Proof of Chain Rule

7. Hint: To prove the "only if" part, choose

$$\varphi(x) = \begin{cases} \dfrac{f(x) - f(c)}{x - c} & \text{if } x \neq c \\ f'(c) & \text{if } x = c \end{cases}$$

8. **Integration**

9. (a) $\sup\{f(x) : x \in [-2, -1]\} = 5$, $\sup\{f(x) : x \in [-1, 1]\} = 4$, $\sup\{f(x) : x \in [1, 4]\} = 2$

$$\therefore U(f, P) = 5(-1 + 2) + 4(1 + 1) + 2(4 - 1) = 19$$

   (b) $L(f, P) = 5$

   (c) $\displaystyle\int_{-2}^{4} f\, dx = 12$

   (d) $5 \leq 12 \leq 19$

   (e) This will depend on the refinement you choose.

10. (a) f is continuous, so it is integrable.

   (b) g is continuous, so it is integrable.

   (c) h is not integrable because the upper sums will always be given by by x^2, while the lower sums will always be given by by $-x^2$. This is similar to the rational indicator function.

   (d) p is not integrable because it is unbounded: the definition of the Riemann integral requires the function be bounded.

   (e) q is integrable because it is piecewise continuous and bounded.

   (f) r is integrable because it is piecewise continuous and bounded.

11. For the partition to be sensible, we need $0 < a < 1$. With the suggested partition, we get $U(f, P) = 1$ and $L(f, P) = 1 - a$. So we have

$$U(f, P) - L(f, P) = 1 - (1 - a) = a$$

   Now choose a value of a that will make that expression less than ϵ.

12.

Improper Integrals

13.

   (a) Improper, infinite interval of integration.

   (b) Improper, infinite discontinuity at $\pi/2$.

   (c) Proper.

   (d) Improper, infinite discontinuity at $\frac{1}{3}$.

   (e) Improper, infinite interval of integration.

   (f) Improper, infinite discontinuity at 0.

   (g) Proper.

   (h) Improper, infinite discontinuity at $\frac{1}{2}$

   (i) Proper.

14.

   (a) 2

   (b) 6

   (c) divergent

   (d) divergent

   (e) 1

   (f) $\frac{\pi}{2}$

   (g) $2\sqrt{5}$

   (h) $\log 2$

7. Series - Selected Answers

Sequence Convergence and Divergence

1. (a) $1, \frac{5}{4}, \frac{49}{36}, \frac{205}{144}, \frac{5125}{3600}$; converges
   (b) $1, \frac{3}{2}, \frac{11}{6}, \frac{25}{12}, \frac{137}{60}$; diverges
   (c) diverges
   (d) converges
   (e) converges
   (f) converges
   (g) converges

2. a_n converges to $\dfrac{2}{3}$ which means the series $\displaystyle\sum_{n=1}^{\infty} a_n$ must diverge.

3. (a) Converges to 5
   (b) Converges to $\dfrac{12}{17}$
   (c) Converges to $\dfrac{1}{12}$

4. Here is what you should compare to (or a multiple of that), and the conclusion.
   (a) $\dfrac{1}{n^2}$, convergent.
   (b) $\dfrac{1}{n}$, divergent.
   (c) $\dfrac{1}{n^{3/2}}$, convergent.
   (d) $\dfrac{1}{4^n}$, convergent.
   (e) $\dfrac{1}{3^n}$, convergent.
   (f) $\dfrac{1}{n^2}$, convergent.
   (g) $\dfrac{1}{n^{2/3}}$, divergent.

5. If $0 \le a_n \le b_n$, then both (A_n) and (B_n) are monotone increasing, and so their convergence can be classified based on boundedness.

6. In the following, $p = \displaystyle\lim_{n\to\infty} \frac{a_{n+1}}{a_n}$.
   (a) $r = e^{-1} < 1$, convergent.
   (b) $r = \frac{1}{2} < 1$, convergent.
   (c) $r = 2 > 1$, divergent.

7. (a) Diverges
   (b) Diverges
   (c) Diverges
   (d) Converges

8. (a) Converges
   (b) Diverges - note that you can't use the alternating series test to conclude this.
   (c) Diverges - note that you can't use the alternating series test to conclude this.
   (d) Converges

9. Yes this is true. Here is an outline of the argument.

   Let $A_n = \displaystyle\sum_{i=1}^{n} a_i$ and $B_n = \displaystyle\sum_{i=1}^{n} b_i$.

   Since both the series are convergent, we have $\displaystyle\lim_{n\to\infty} A_n = L$ and $\displaystyle\lim_{n\to\infty} B_n = M$.

   Now consider the partial sum

$$\begin{aligned} S_n &= \sum_{i=1}^{n}(a_i + b_i) \\ &= \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i \\ &= A_n + B_n \\ \text{Hence } \lim_{n\to\infty} S_n &= \lim_{n\to\infty}(A_n + B_n) \\ &= \lim_{n\to\infty} A_n + \lim_{n\to\infty} B_n \\ &= L + M \end{aligned}$$

Since the limit exists, the partial sums converge, and hence $\sum_{n=1}^{\infty}(a_n + b_n)$ converges.

10. This one is actually false!

    To help you find a counterexample, choose
    $$a_n = \frac{(-1)^n}{\sqrt{n}}$$
    Now you pick b_n.

Power Series

11. (a) $[-1, 1]$, $\rho = 1$
    (b) $(-2, 2)$, $\rho = 2$
    (c) $\mathbb{R}$
    (d) $(-\frac{1}{3}, \frac{1}{3}]$, $\rho = \frac{1}{3}$
    (e) $[-1, 1)$, $\rho = 1$

Taylor Series and Power Series

12. (a) $\sum_{n=0}^{\infty}(-1)^n \frac{(2x)^{2n+1}}{(2n+1)!}$ converges for $x \in \mathbb{R}$
    (b) $\sum_{n=0}^{\infty}(-1)^n \frac{x^{2n}}{(2n)!}$ converges for $x \in \mathbb{R}$
    (c) $\sum_{n=0}^{\infty}(-1)^n x^n$ converges for $x \in (-1, 1)$
    (d) $\sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$ converges for $x \in \mathbb{R}$
    (e) $\sum_{n=1}^{\infty}(-1)^{n+1} \frac{(2x)^n}{n}$ converges for $x \in (-\frac{1}{2}, \frac{1}{2}]$
    (f) $\sum_{n=0}^{\infty}(-1)^n (n+1) x^n$ converges for $x \in (-1, 1)$

13. (a)
    $$\begin{aligned}
    &= \sin(\pi/4) + \cos(\pi/4)(x - \pi/4) - \frac{\sin(\pi/4)}{2!}(x - \pi/4)^2 - \frac{\cos(\pi/4)}{3!}(x - \pi/4)^3 + \frac{\sin(\pi/4)}{4!}(x - \pi/4)^4 + \dots \\
    &= (1/\sqrt{2}) + (1/\sqrt{2})(x - \pi/4) - \frac{1/\sqrt{2}}{2!}(x - \pi/4)^2 - \frac{1/\sqrt{2}}{3!}(x - \pi/4)^3 + \frac{1/\sqrt{2}}{4!}(x - \pi/4)^4 + \dots \\
    &= (1/\sqrt{2}) + (1/\sqrt{2})(x - \pi/4) - \frac{1}{(\sqrt{2})2!}(x - \pi/4)^2 - \frac{1}{(\sqrt{2})3!}(x - \pi/4)^3 + \frac{1}{(\sqrt{2})4!}(x - \pi/4)^4 + \dots
    \end{aligned}$$
    (b) $\sum_{n=0}^{\infty} \frac{(-1)^n (x-2)^n}{2^{n+1}}$
    (c) $e^{-3} \sum_{n=0}^{\infty} \frac{(x+3)^n}{n!}$

14. (a) $P_n(x) = \sum_{k=0}^{n}(-1)^k x^k$, $R_n(x) = \frac{(-1)^{n+1} x^{n+1}}{(1+c)^{n+2}}$
    (b) With $0 < x < 1$ and $0 < c < x$ we have
    $$\begin{aligned}
    |R_n(x)| &= \left| \frac{(-1)^{n+1} x^{n+1}}{(1+c)^{n+2}} \right| \\
    &= \frac{|x|^{n+1}}{|1+c|^{n+2}} \\
    &\leq \frac{|x|^{n+1}}{|1+0|^{n+2}} \\
    &= |x|^{n+1} \to 0 \text{ as } n \to \infty
    \end{aligned}$$
    (c) Finally, take the limit as $n \to \infty$ of
    $$f(x) = P_n(x) + R_n(x)$$

15. (a) $17/18 \approx 0.9444$ (b) $1703/1800 \approx 0.94611$

16.